

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	D	F	H	S_1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}F_{r_1}^T C F_{r_1} M_F$	$+\overline{\lambda_{T3}}S_{2,j_2}\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	
								$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$+\overline{L_{a_1,i_1}}P_R F_{r_1}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y^1,a_1,j_1}$ $+\overline{y_{2,i_1,r_1}}F_{r_1}^T C P_L L_{a_1,i_1}S_{2,j_1}\text{Bar}(T3Y2CG)_{a_1,j_1}$ $+\overline{y_{1,i_1,r_1}}F_{r_1}^T C P_L L_{a_1,i_1}\text{Bar}(T3Y1CG)_{a_1,j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}P_R F_{r_1}y_{2,i_1,r_1}\mathcal{I}_{y^2,a_1,j_1}S_{2,j_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^2$ $+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1\lambda_{H2}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2\lambda_{H2}^2$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}F_{j_1,r_1}^T C F_{j_1,r_1} M_F$	$+\overline{\lambda_{T3}}S_{2,j_2}\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	
								$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$+\overline{L_{a_1,i_1}}P_R F_{j_1,r_1}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{2,i_1,r_1}}F_{j_1,r_1}^T C P_L L_{a_1,i_1}S_{2,j_1}\text{Bar}(T3Y2CG)_{a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}}F_{j_1,r_1}^T C P_L L_{a_1,i_1}\text{Bar}(T3Y1CG)_{a_1,j_1,j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}P_R F_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y^2,a_1,j_1,j_1}S_{2,j_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^2$ $+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1\lambda_{H2}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2\lambda_{H2}^2$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}}F_{j_1,r_1}M_F$	$+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1\lambda_{H2}^1$	$-S_1S_1^\dagger M_{S_1}^2$ $+S_1S_{2,j_2}S_1^\dagger S_{2,j_1}^\dagger\mathcal{I}_{12,j_1,j_2}^1\lambda_{12}^1$ $+\frac{1}{2}S_1^2(S_1^\dagger)^2\lambda_{S_1}^1$
									$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}P_L L_{a_1,i_1}S_{2,j_1}\text{Bar}(T3Y2CG)_{a_1,j_1,j_1}$ $+\overline{L_{a_1,i_1}}P_R F_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y^2,a_1,j_1,j_1}S_{2,j_1}^\dagger$	$+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1\lambda_{H2}^1$ $+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2\lambda_{H2}^2$ $+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2\lambda_{H2}^2$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\overline{F_{j_1,r_1}}F_{j_1,r_1}M_F$	$+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^1$	
									$+\overline{L_{a_1,i_1}}C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^T C P_L F_{j_1,r_1}\text{Bar}(T3Y1CG)_{a_1,j_1,j_1}S_{1,j_1}^\dagger$	$+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}^1\lambda_{H1}^1$ $+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^2$ $+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2\lambda_{H1}^2$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\overline{F_{j_1,r_1}}F_{j_1,r_1}M_F$	$+\overline{\lambda_{T3}}S_{2,j_2}\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	
								$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$+\overline{L_{a_1,i_1}}C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}P_L L_{a_1,i_1}S_{2,j_1}\text{Bar}(T3Y2CG)_{a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}}L_{a_1,i_1}^T C P_L F_{j_1,r_1}\text{Bar}(T3Y1CG)_{a_1,j_1,j_1}S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}}P_R F_{j_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y^2,a_1,j_1,j_1}S_{2,j_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger$ $+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1\lambda_{H1}^1$ $+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2\lambda_{H1}^2$ $+\frac{1}{2}H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2\lambda_{H1}^2$ $+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1\lambda_{H2}^1$ $+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2\lambda_{H2}^2$ $+\frac{1}{2}H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2\lambda_{H2}^2$	

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	F	D	S_1	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \end{aligned}$	$+\frac{1}{2} i F_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \end{aligned}$	$+\frac{1}{2} i F_{j_1,r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^2 \lambda_{S2}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$		
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & -M_{S_2}^2 S_2 S_2^\dagger \\ & +S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1 \\ & +\frac{1}{2} S_2^2 (S_2^\dagger)^2 \lambda_{S2}^1 \end{aligned}$	$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^2 \lambda_{S1}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^2 \lambda_{S1}^2 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^2 \lambda_{S2}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	H	S_2	D	S_2	F
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0						
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0						
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{\gamma^1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} L_{a_1,i_1}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1} S_1^\dagger$	$+\overline{\lambda_{T3}} S_{2,j_1} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}^1 \lambda_{H1}^1$				
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$			$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$+\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1,j_1}$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{\gamma^2,a_1,j_1} S_2^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$						

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+H_{a_1} H_{a_2} S_{1,j_1} \mathcal{L}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger$ $+ \overline{\lambda_{T3}} S_2 \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1 \lambda_{H2}{}^1$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	